

Growth Facts

- Kaldor (1961) established a set of stylized facts about the growth process.
- Fact 1: There is secular growth in per-capita output. Growth rate does not tend to diminish.

Growth Facts

- Fact 2: Physical capital per worker grows over time.
- Fact 3: Capital to output ratio is relatively constant over time.
- Fact 4: Rate of return to capital is trendless.
- Fact 5: Capital and labor shares are constant over time.

Growth Facts

- Fact 6: There is substantial variation in per-capita output growth rates across countries.

Twin Fact: output per capita differs substantially across countries.

- Task of growth theory is to develop models that are consistent with these stylized facts.

US Data: the long view

- We now discuss some evidence on the growth facts for the United States for the past decades.
- We start on data on aggregate output (GDP).
- What is GDP?

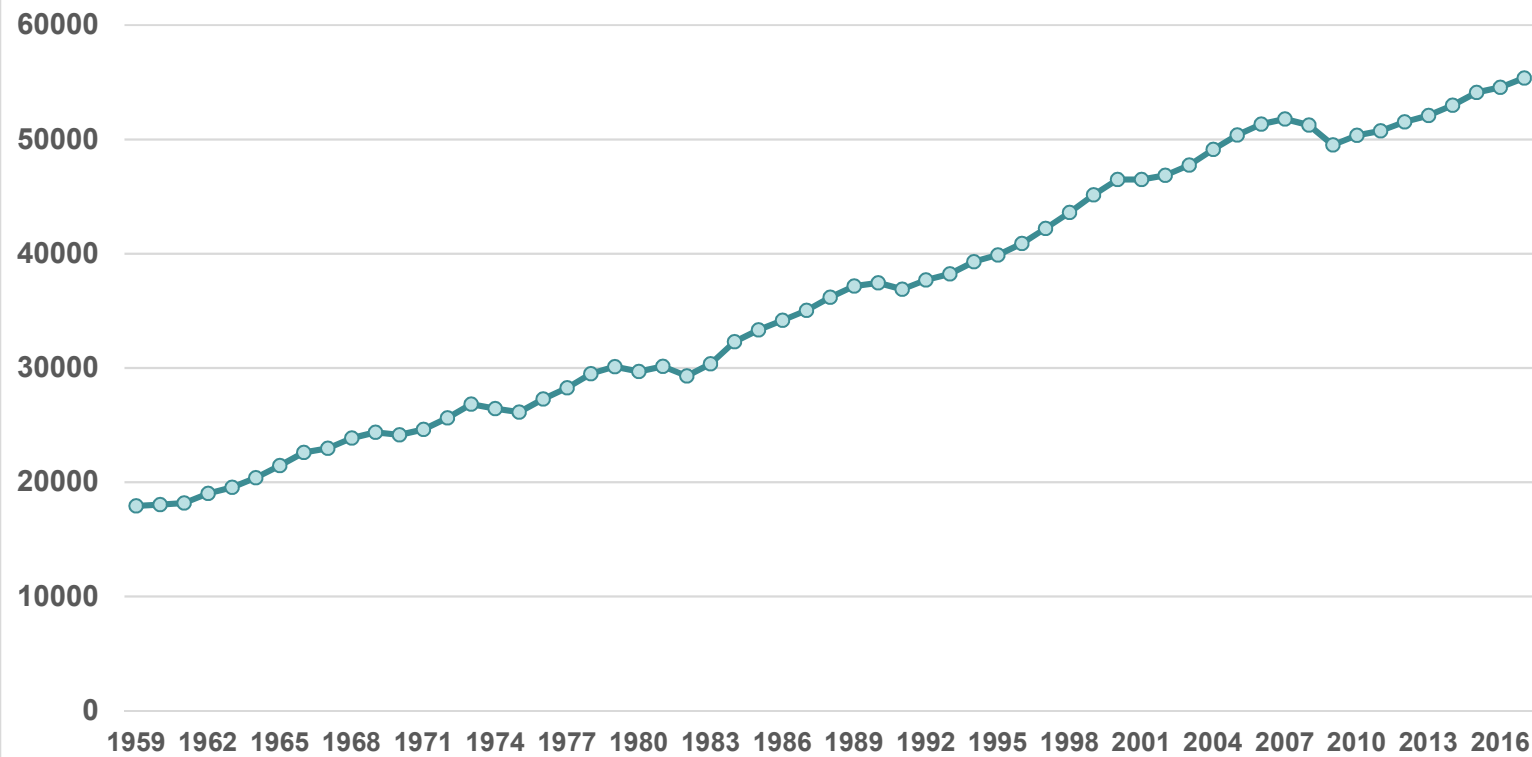
Gross Domestic Product (GDP)

- Definition: GDP is the market value of all final goods and services produced by an economy at a point in time.

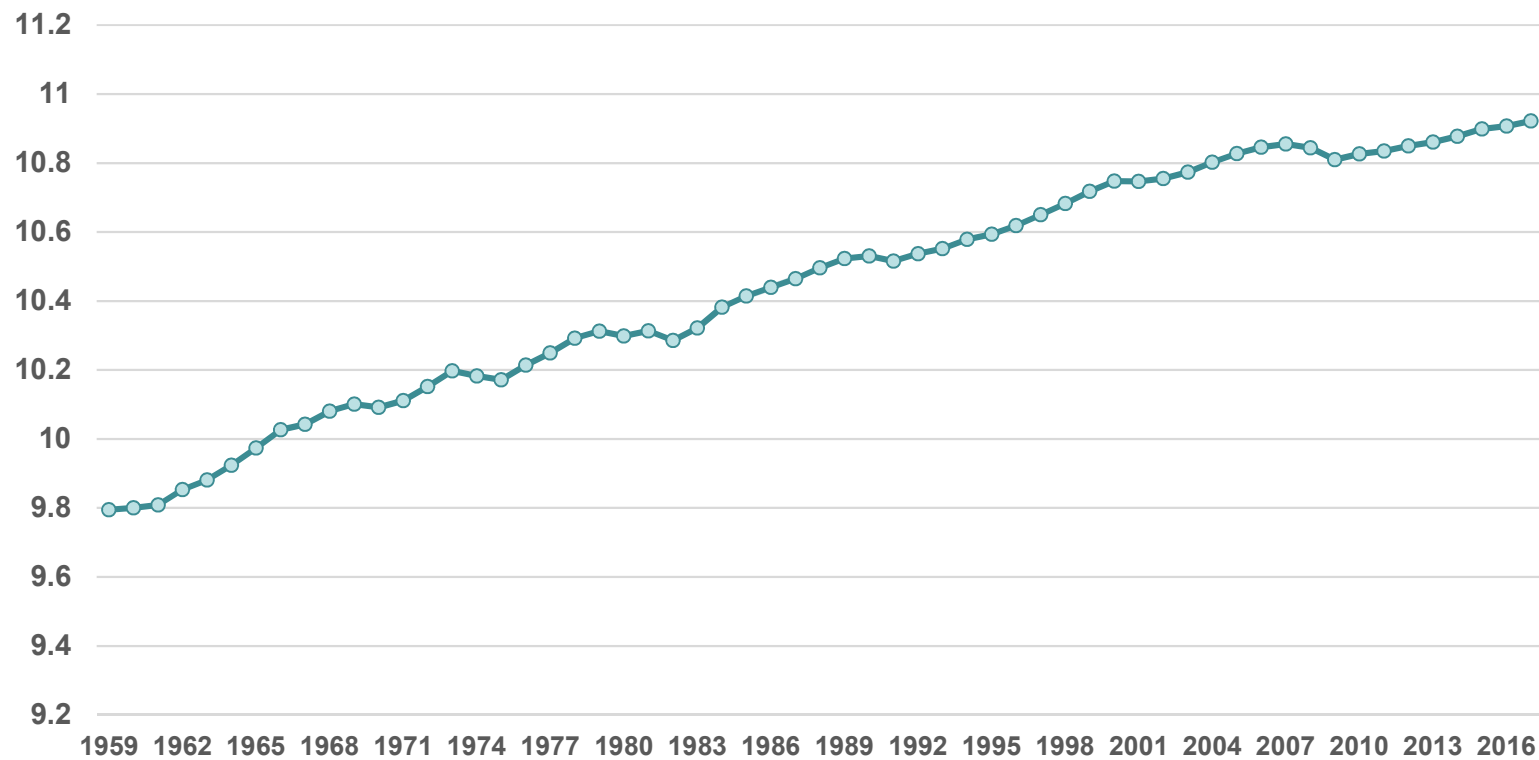
Gross Domestic Product (GDP)

- Definition (2): Sum of the **value added** of all sectors in the economy at a point in time.
- Definition (3): Sum of **compensation** to all factors of production at a point in time.
- All notions theoretically coincide. **What is GNP?**

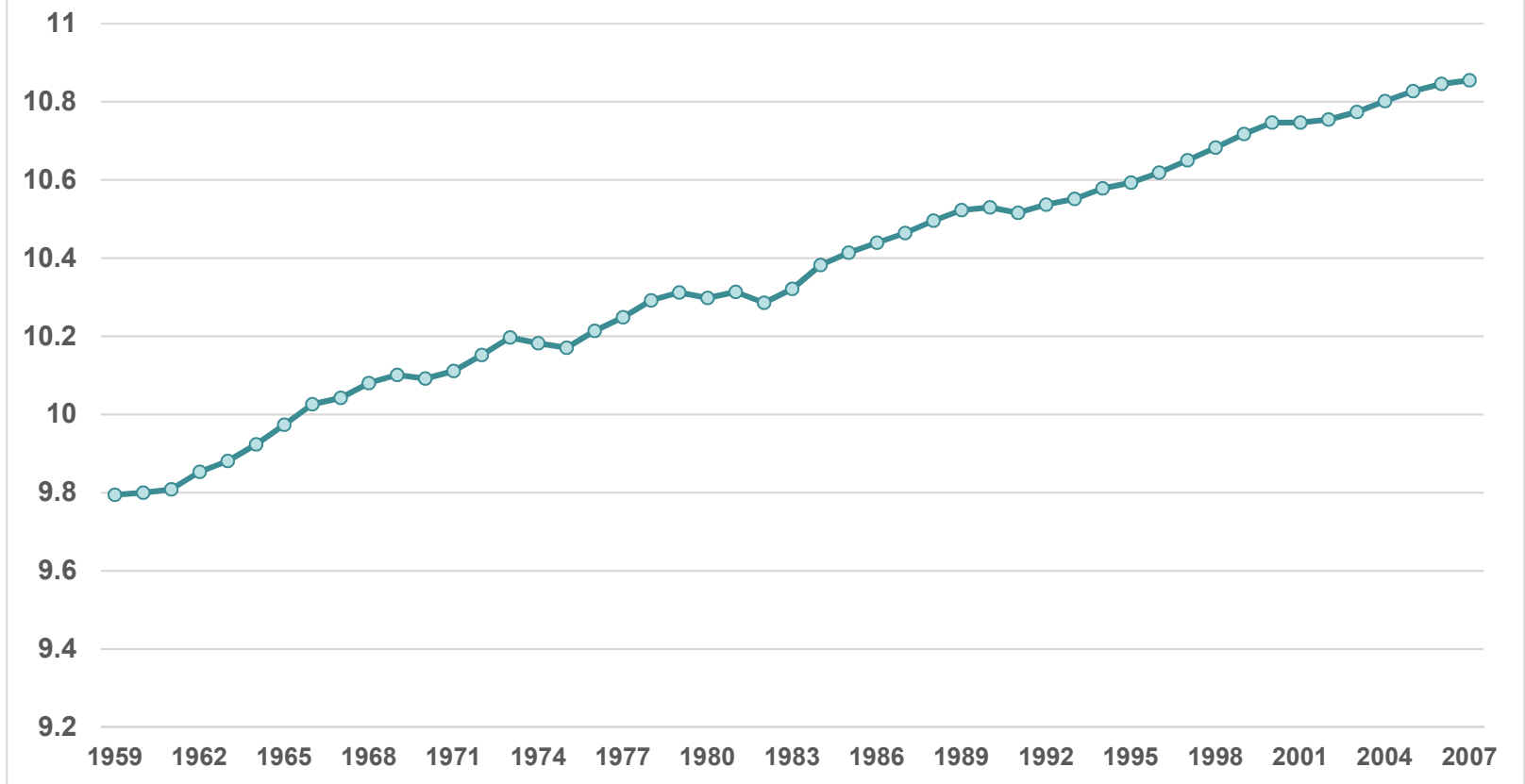
GDP per capita -- 1959-2017



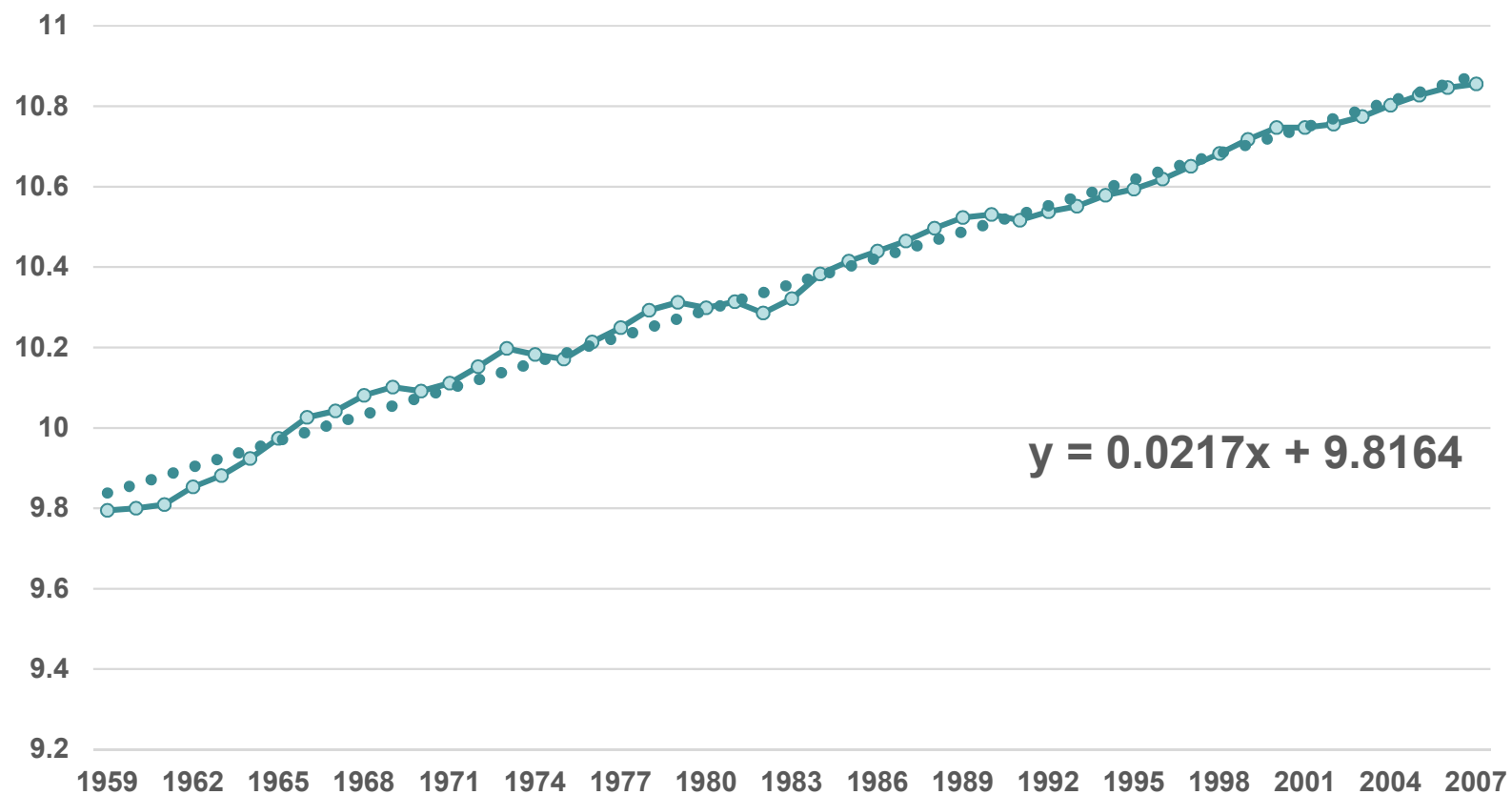
GDP per capita (log) -- 1959-2017



GDP per capita (log) -- 1959-2007

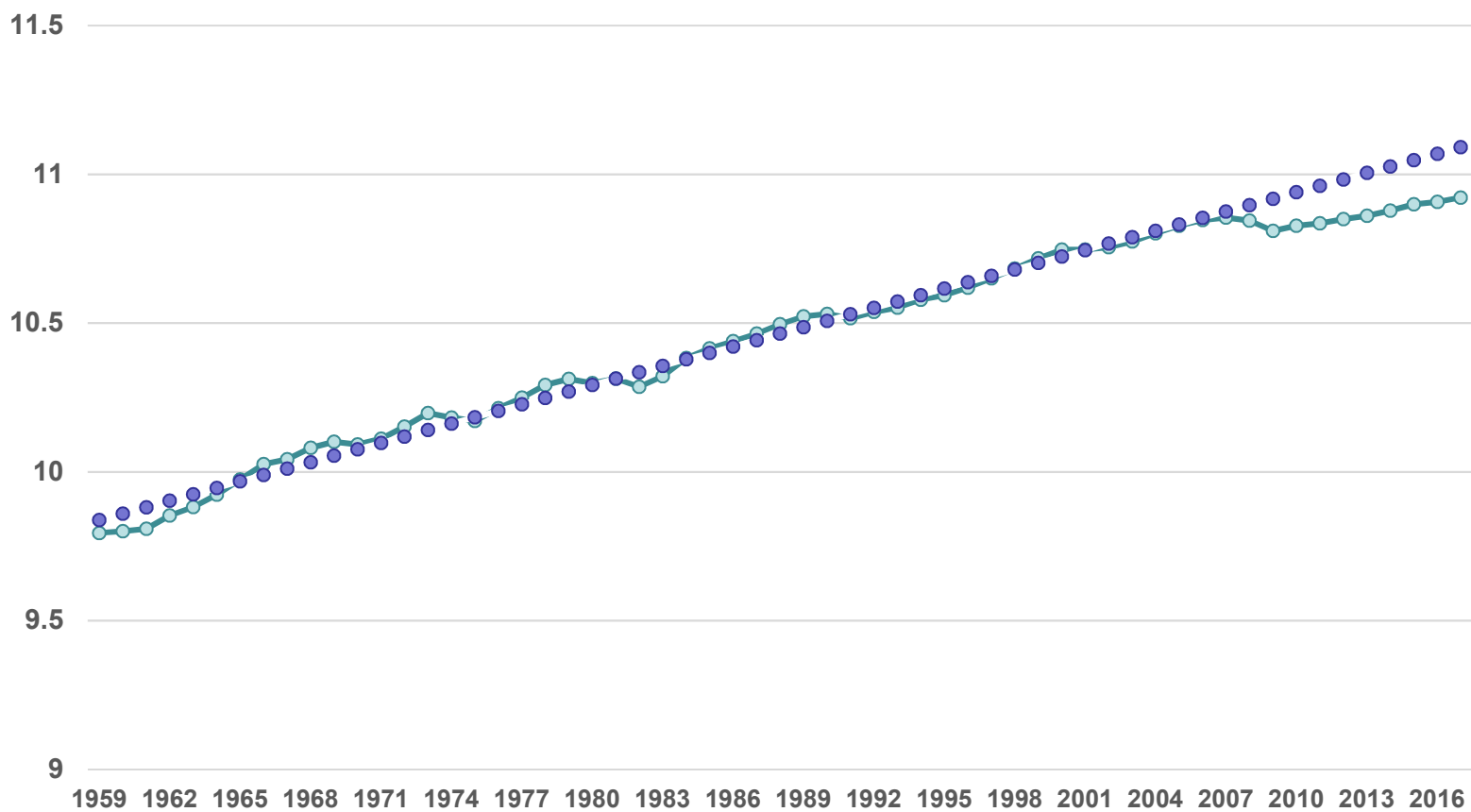


GDP per capita (log) -- 1959-2007



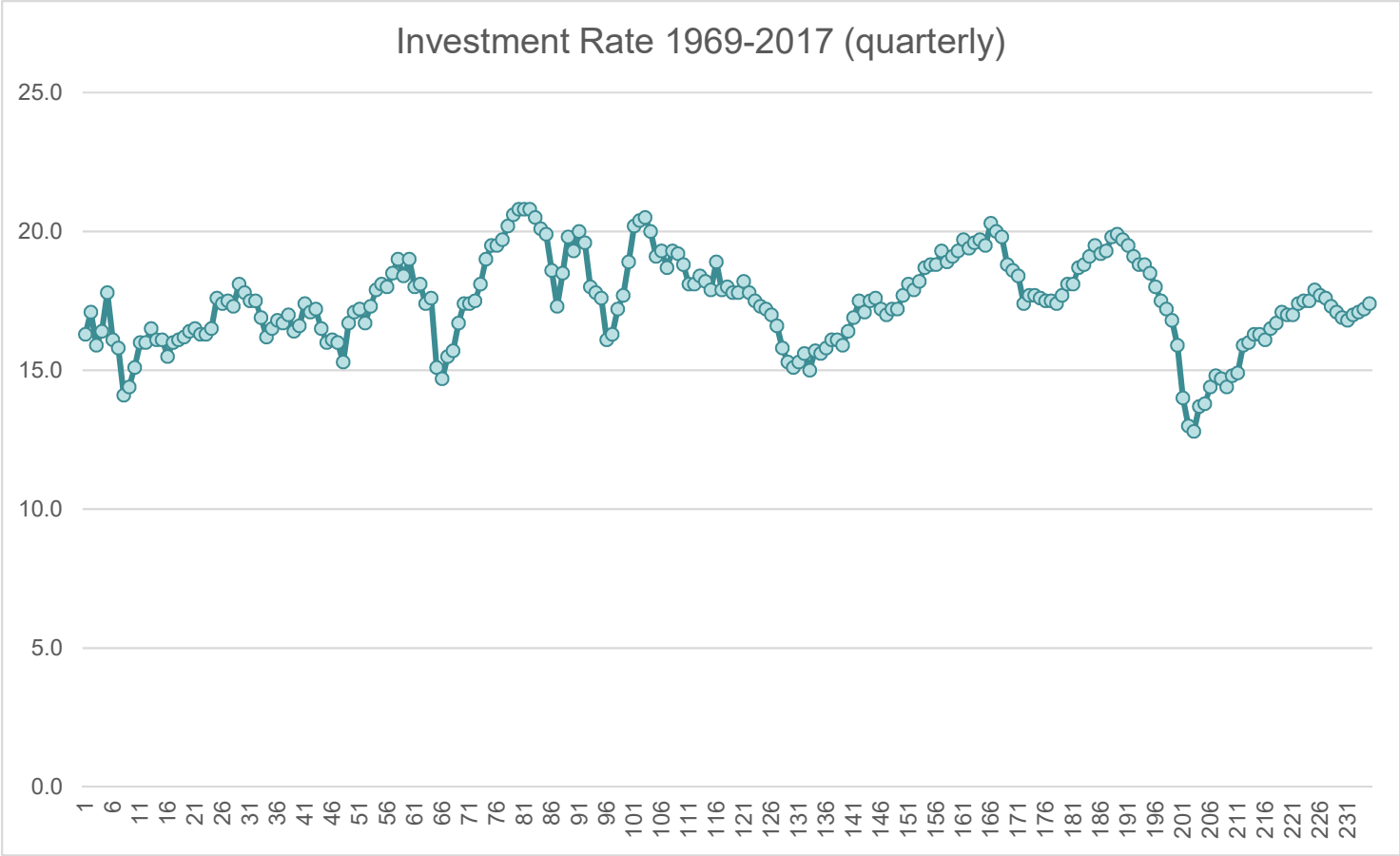
- Slope of linear trend approximates growth rate. **Why?**
- Growth rate is about 2% per year.
- This has drastic implications. **Output per capita doubles in about 35 years.**
- What is the current value for US GDP per capita?

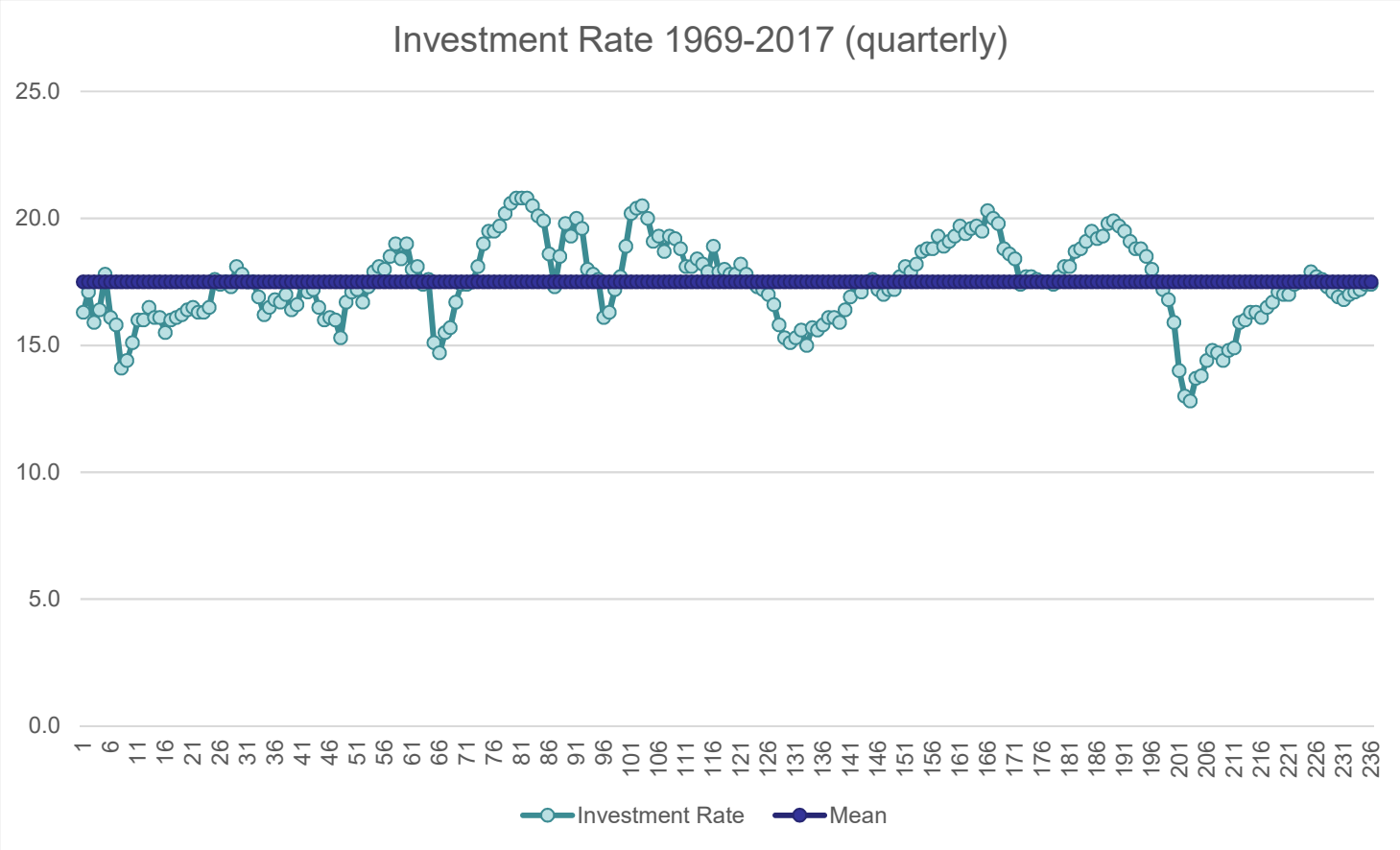
GDP per capita (log) -- 1959-2017

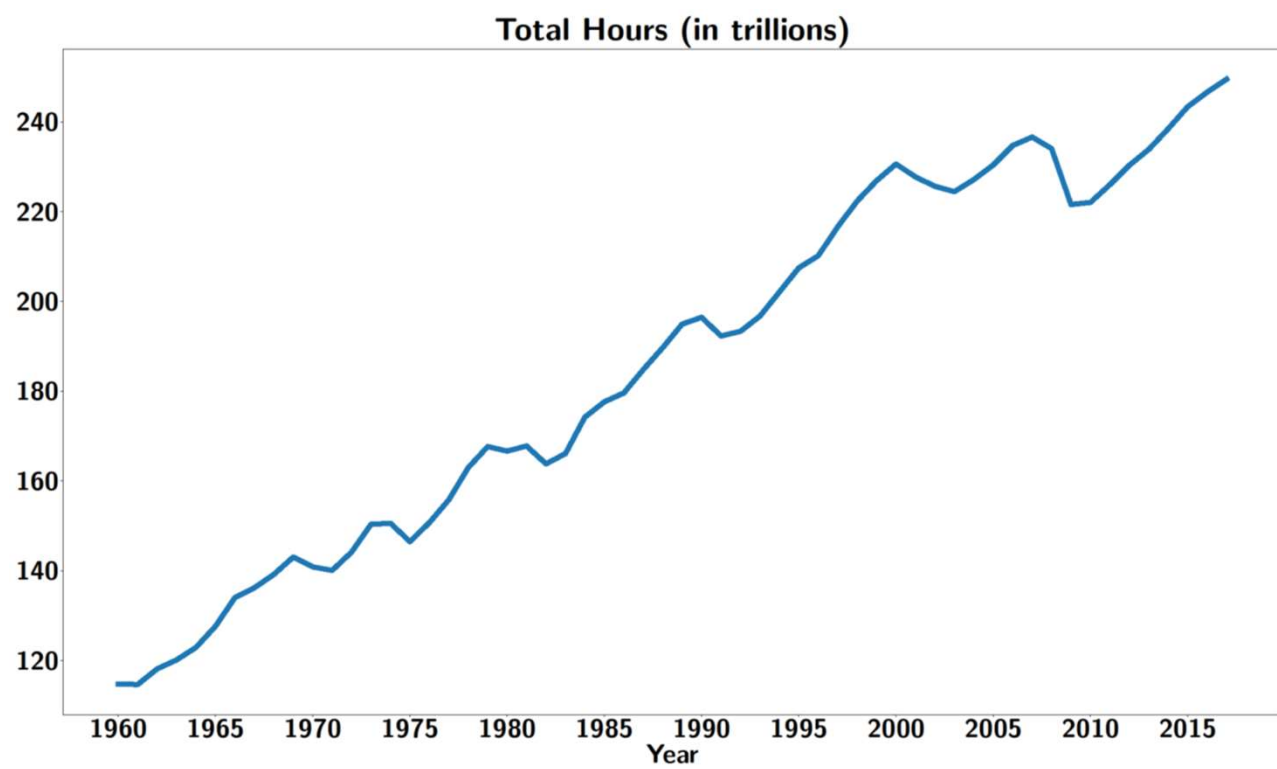


Related Aggregate Data

- **Big picture:**
- (i) investment rate is **trendless**;
- (ii) hours worked grow over time;
- (iii) hours worked per adult are **roughly constant over time**;
- (iv) employment rates are about constant over time (but differ by gender).

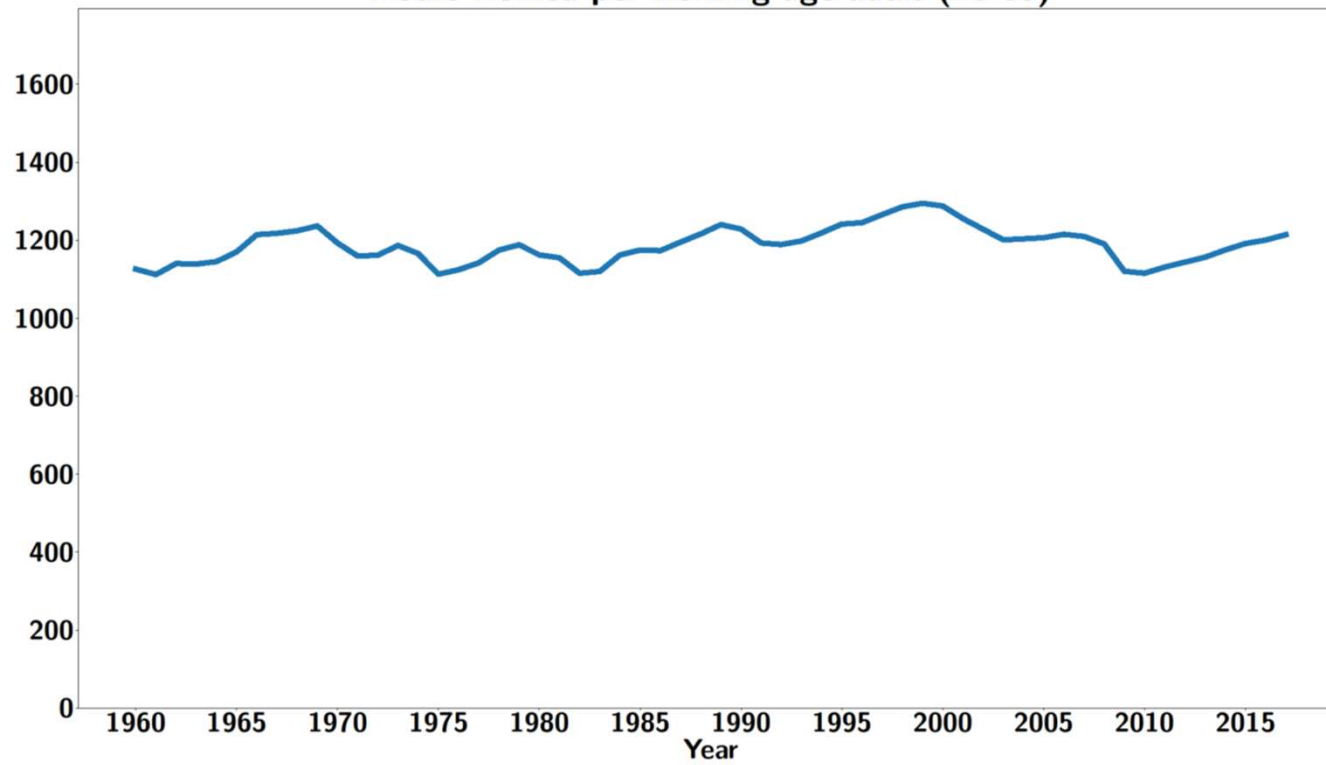




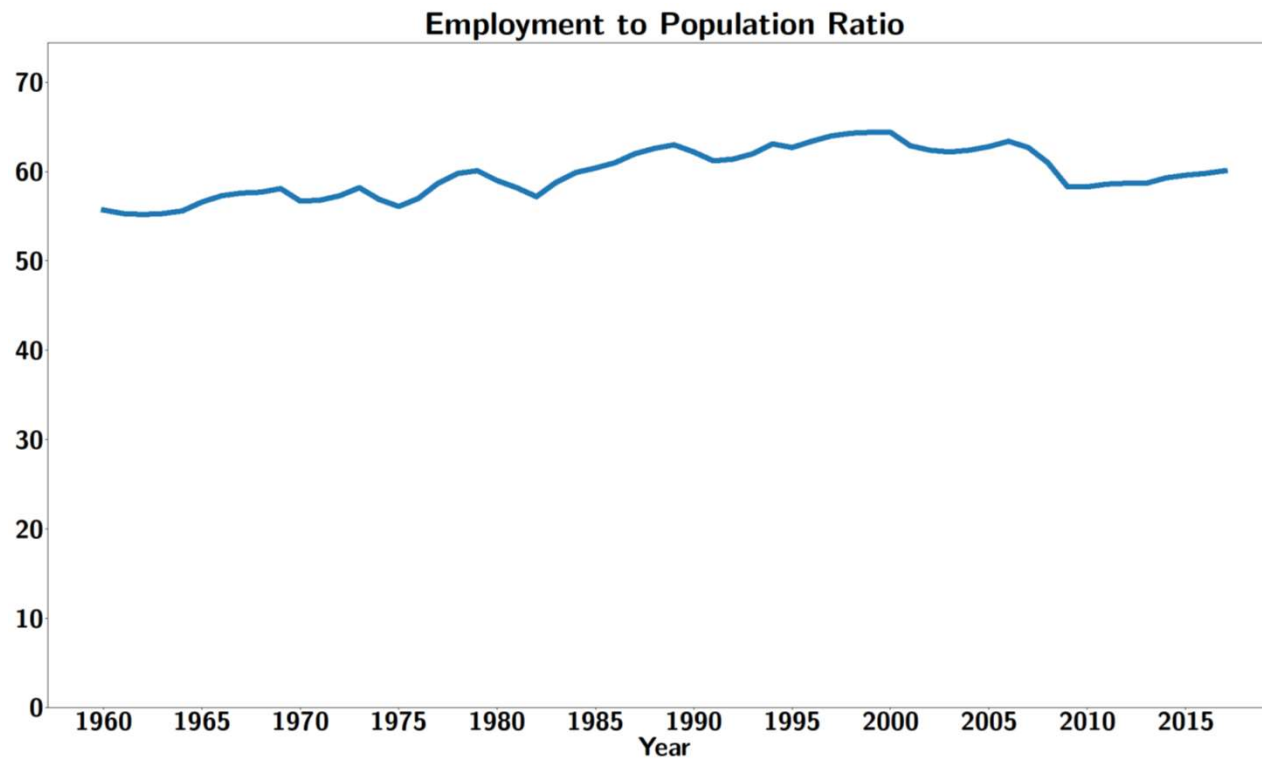


- Total hours reflect the number of hours worked by those employed (**intensive margin**), and the number of persons employed (**extensive margin**).

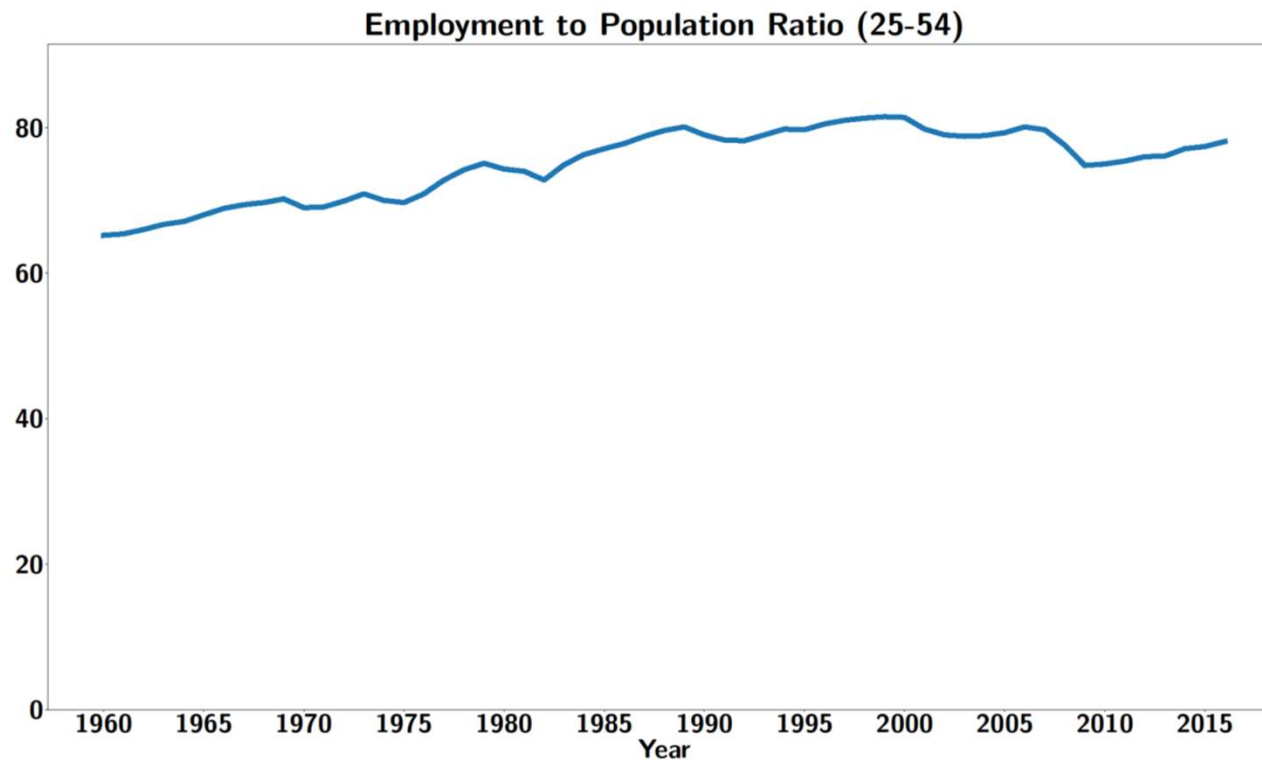
Hours worked per working age adult (16-65)



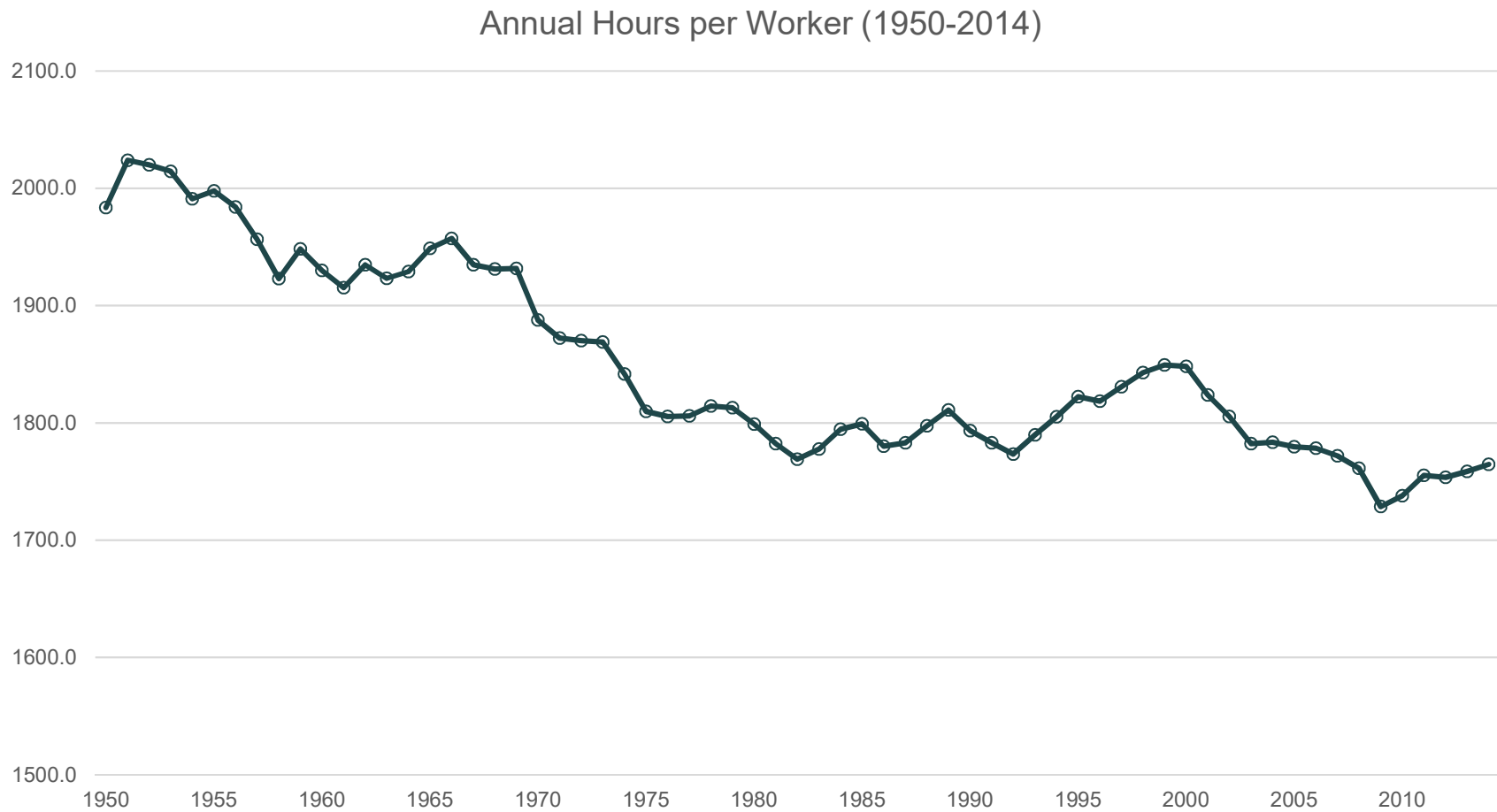
Extensive Margin (I)

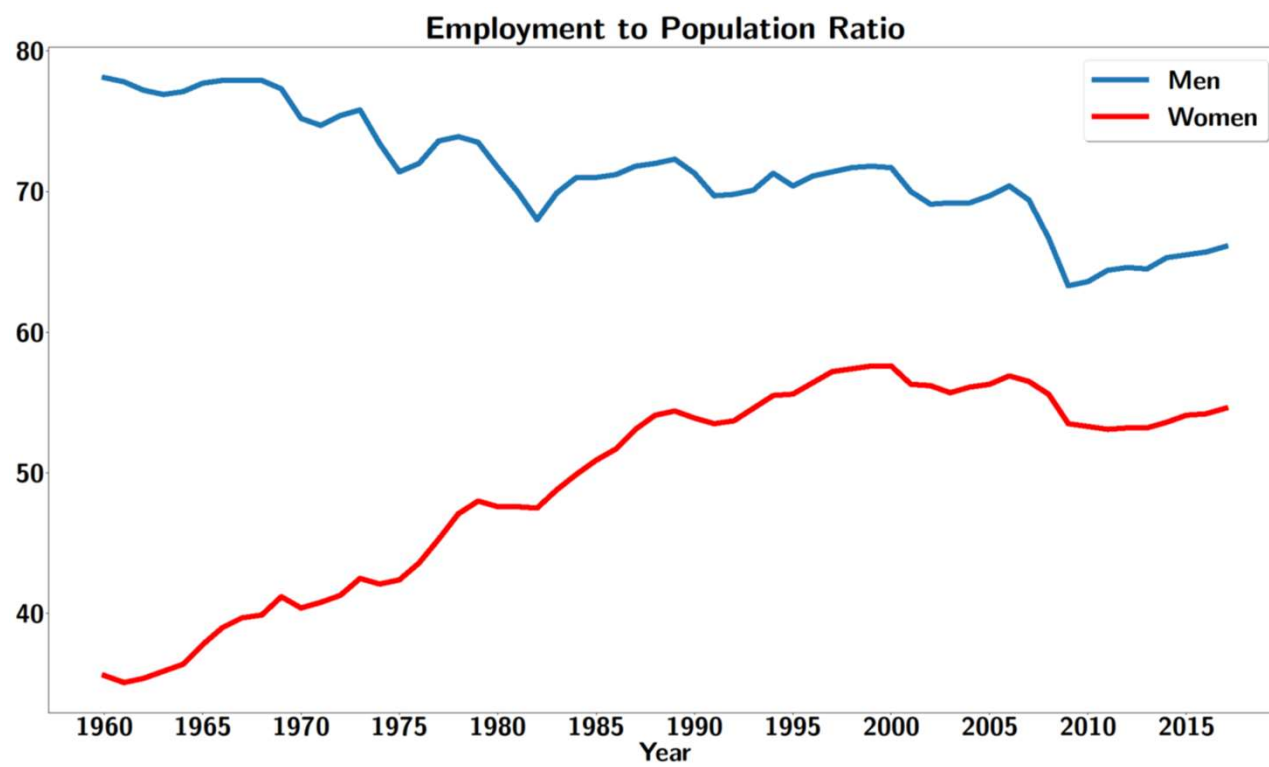


Extensive Margin (II)



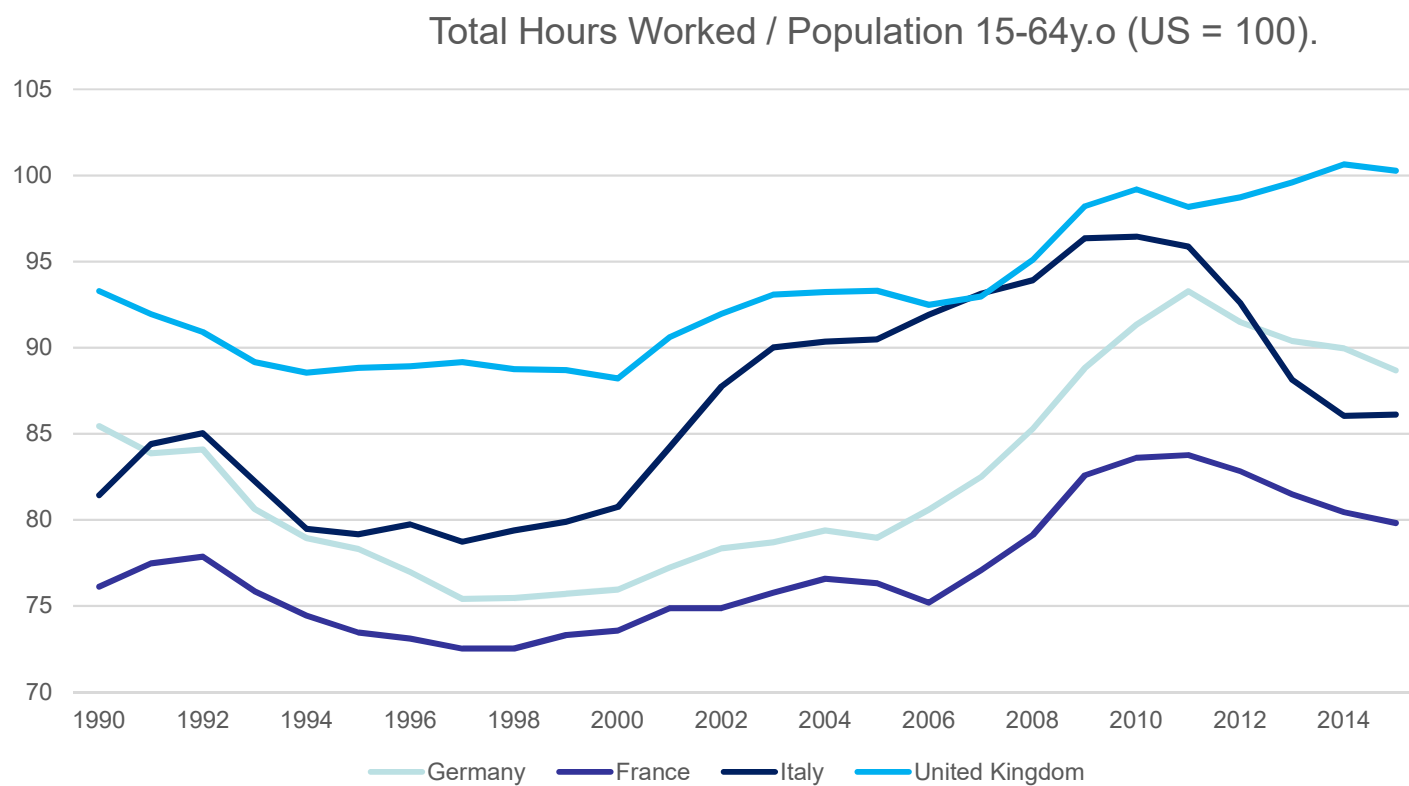
Intensive Margin in Hours Worked



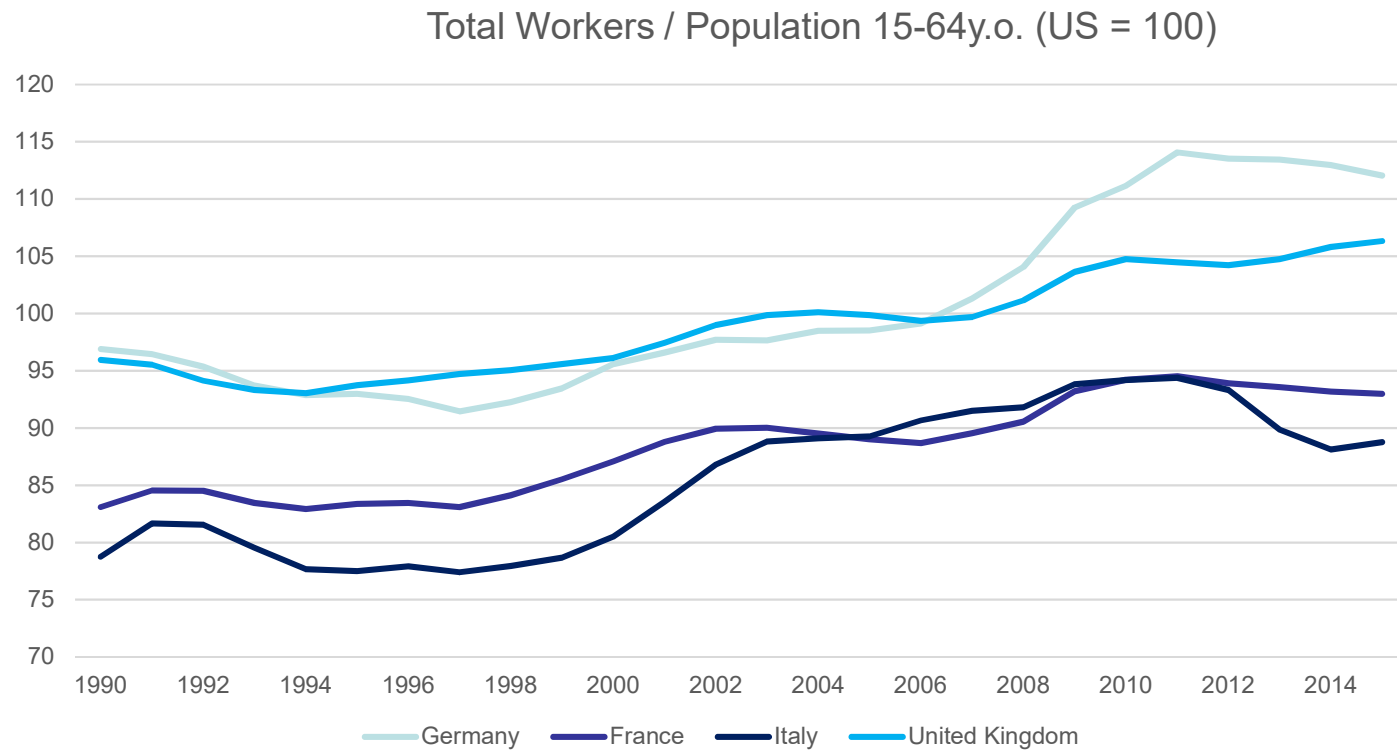


- How does the US relate to other countries in terms of levels of hours worked?

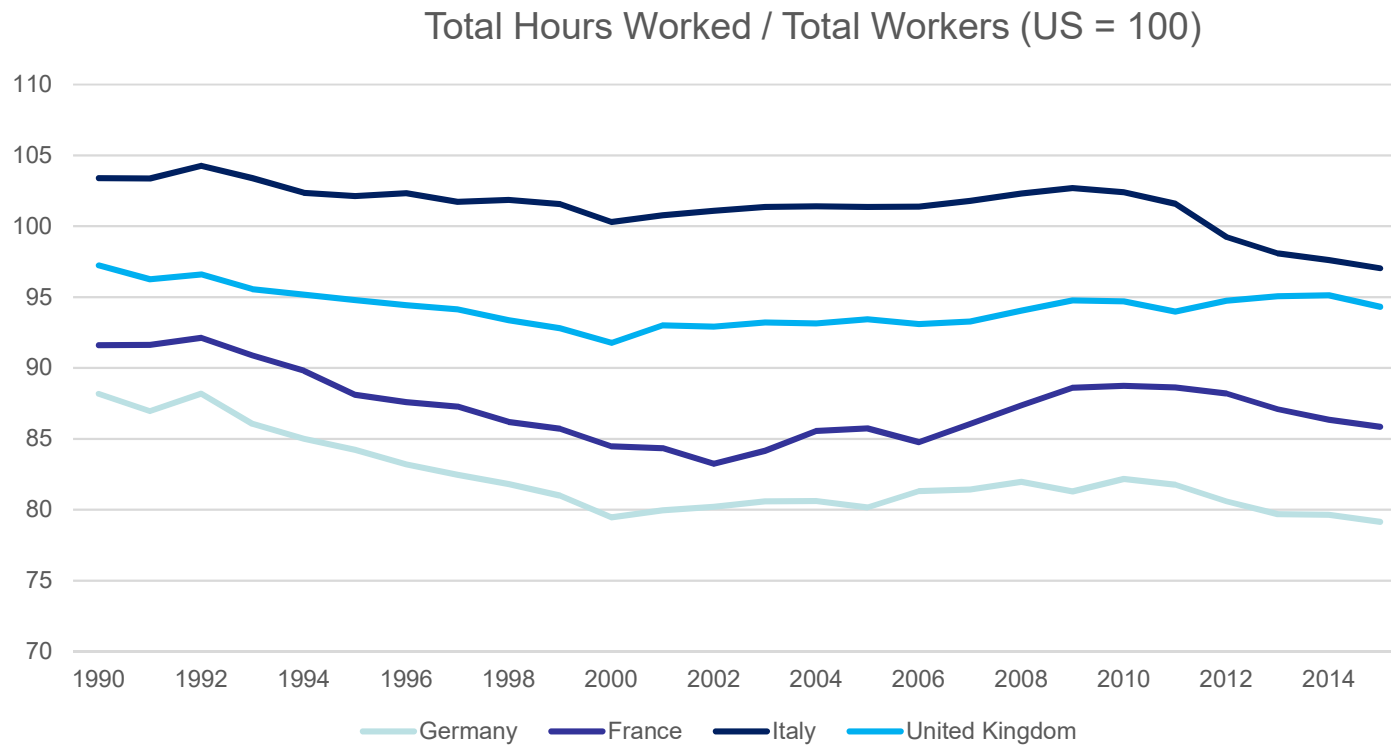
Hours Worked per Adult – U.S. vs European Countries



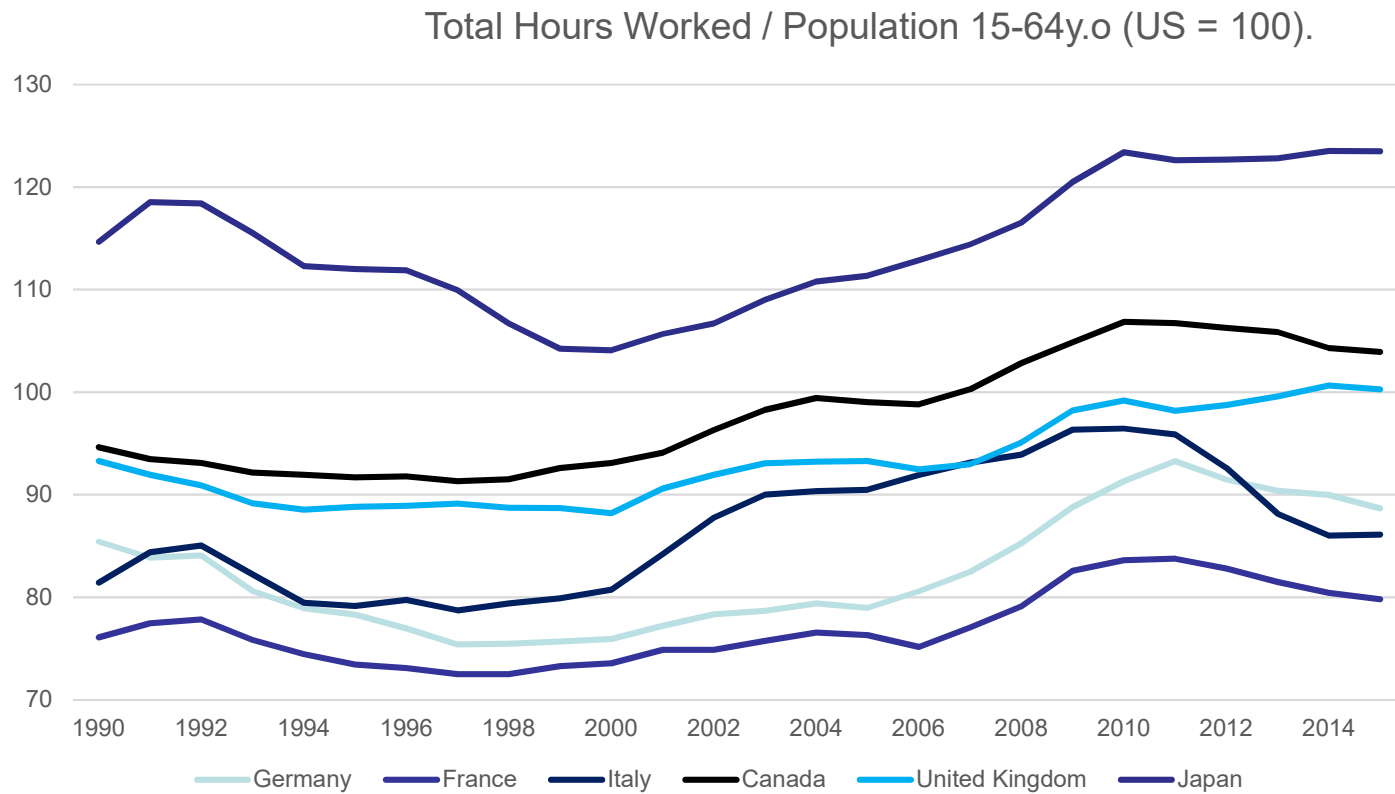
Employment Rates – U.S. vs European Countries



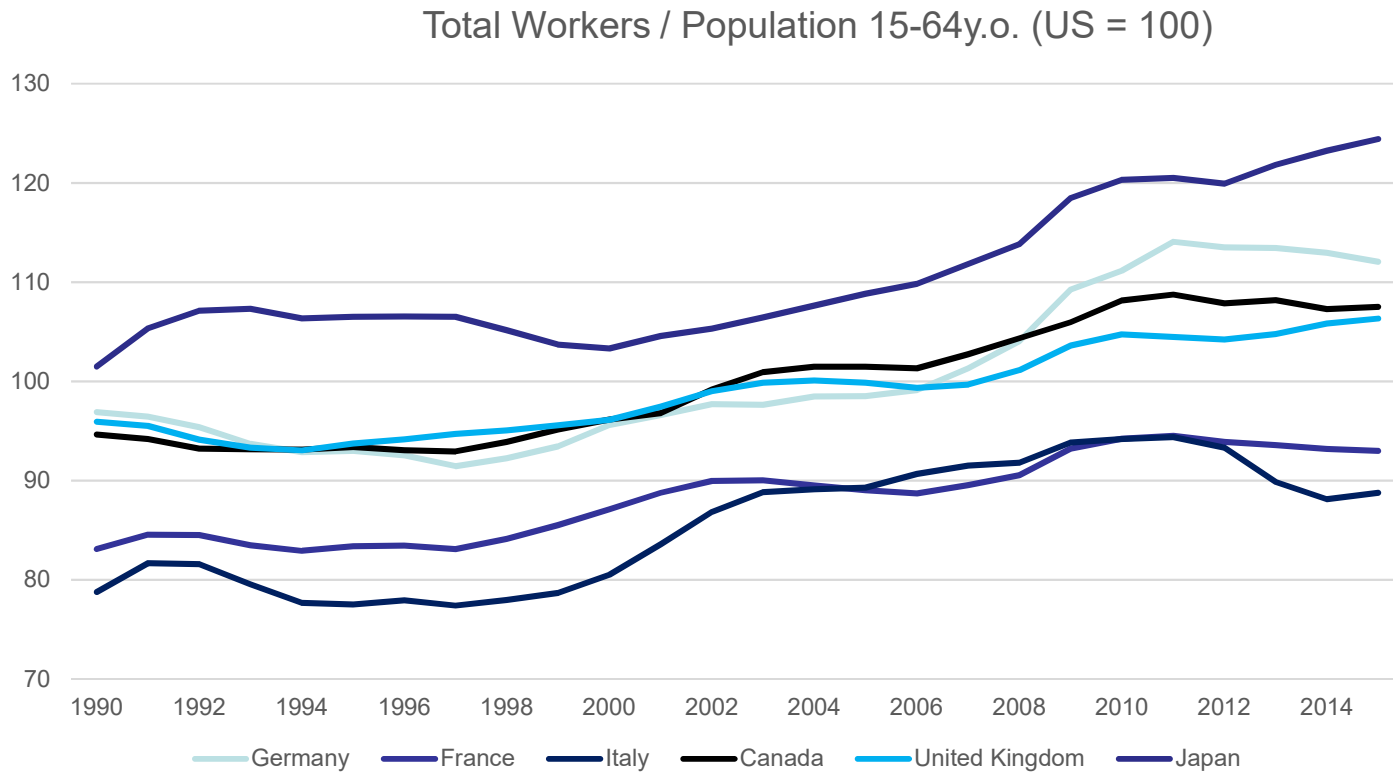
Hours Worked per Worker – U.S. vs European Countries



Hours Worked per Adult – US vs G7 Countries



Employment Rates – U.S. vs G7 Countries



Hours Worked per Worker – U.S. vs G7 Countries

Total Hours Worked / Total Workers (US = 100)

